Notes: Hirshleifer (JF, 2020)

Presidential Address: Social Transmission Bias in Economics and Finance

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1 Big picture

- **Question.** What happens to economic beliefs and behavior when ideas and signals pass from person to person with systematic distortion or selective transmission?
- **Setting.** This presidential address is a conceptual and theoretical paper. It introduces *social economics and finance* and develops five "fables" of social transmission bias.
- **Core challenge.** Standard finance typically emphasizes prices, private information, and individual bias, but underplays the fact that people talk to each other, observe each other, and transmit information imperfectly.
- **Main conceptual contribution.** Hirshleifer proposes *social transmission bias* as a core building block: signals or ideas are systematically shifted, simplified, censored, or selectively repeated as they move through social contact.
- **Two main forms of transmission bias.** The first is *distortion* (what gets transmitted is directionally modified). The second is *selection* (which messages get transmitted depends on the underlying signal or outcome).
- **Main new model.** The paper's most formal contribution is **Fable 4**, a model of biased information percolation in which agents exchange biased signal summaries and misperceive how much information has already flowed through the network.
- **Main theoretical result.** Small message biases can compound recursively and generate large temporary departures from fundamentals: action booms, asset price bubbles, peak oscillation near the top, and eventual correction when public information becomes sufficiently precise.
- **Main behavioral insight.** Socially emergent outcomes can mimic direct psychological biases. Investors can look as if they love lottery-like payoffs, consume as if present-biased, or extrapolate returns, even when no such primitive taste or belief distortion is built in.
- **Main asset-pricing implications.** The biased percolation model generates bubble-like price paths, event-date reversals followed by continuation, short-horizon momentum, longer-horizon reversal, and belief changes that look like extrapolation.
- **Main empirical implication.** Researchers should study not only investor preferences and beliefs, but also social/network proxies, communication technologies, visibility, sender incentives, and textual or narrative features that create transmission bias.
- **Economic takeaway.** Finance should not stop at "what signals exist" or "what biases individuals have." It must also ask *how information is socially transmitted.* Once that is modeled, many economic and financial phenomena look different.

2 Model environment and key objects

2.1 Social economics and finance

The paper defines **social economics and finance** as the study of the social processes that shape economic thinking and behavior. The key idea is that people observe each other and talk to each other, where "talking" includes conversation, written communication, blogs, and social media.

Interpretation. The paper is not rejecting information economics or behavioral finance. It is adding a missing layer: beliefs and actions are shaped not only by private signals and personal bias, but also by the way messages travel through a social system.

2.2 Social transmission bias

A **social transmission bias** is a systematic directional shift in ideas or signals induced by social transactions.

The paper emphasizes two broad forms.

- **Signal distortion.** A message is transmitted with altered sign, intensity, or emphasis.
- **Selection.** Some messages are more likely to be transmitted than others, so receivers observe a biased sample.

Interpretation. Distortion says "the message changes in transit." Selection says "what gets transmitted is nonrandom." Both create biased social learning if receivers fail to adjust.

2.3 The five fables

The address is organized around five stylized mechanisms.

1. **Bandwidth constraints and simplistic thinking.** Communication limits strip away nuance, so iterated discussion can push beliefs toward oversimplified extremes.
2. **Self-enhancing transmission bias.** Investors talk more about good outcomes than bad ones, causing high-variance or high-skewness strategies to spread even when inferior.
3. **Visibility bias and overconsumption.** Consumption is more visible than nonconsumption, so people infer that others are consuming more than is really justified, which can induce undersaving.
4. **Biased information percolation.** Signals are repeatedly shared with bias and the extent of social learning is misperceived, producing action booms and price bubbles.
5. **Biased transmission of folk models.** Narratives and simplified worldviews spread contagiously, and their population dynamics can shape prices and investor behavior.

Interpretation. The paper's method is broad on purpose. It shows that many seemingly unrelated behaviors can be understood as consequences of transmission bias.

2.4 Key objects in Fable 4: common payoff, signals, and actions

The most formal model studies a common payoff X and agents who repeatedly exchange information about it. The payoff is Gaussian:

$$X \sim N\left(\bar{X}, \frac{1}{\tau^X}\right). \tag{1}$$

Each agent i initially receives one private signal,

$$S^i = X + \epsilon^i, \qquad \epsilon^i \sim N\left(0, \frac{1}{\tau^S}\right), \tag{2}$$

and at discrete dates $t = 0, 1, \ldots, T-1$ everyone observes public signals

$$S_t^P = X + \epsilon_t^P, \qquad \epsilon_t^P \sim N\left(0, \frac{1}{\tau_t^P}\right). \tag{3}$$

The cumulative precision of public information grows as

$$\tau_s^Z = e^{\zeta \lfloor s \rfloor} \tau^P, \qquad 0 \le s < T. \tag{4}$$

At continuous date s, agent i chooses an action intensity θ_s^i. With CARA utility and Gaussian uncertainty, the optimal action takes the mean-variance form

$$\theta_s^i = \frac{\mathbb{E}[X \mid \mathcal{I}_s^i] - v_s}{\gamma \mathrm{Var}(X \mid \mathcal{I}_s^i)}. \tag{5}$$

Interpretation. This is a standard rational benchmark with one nonstandard ingredient: the information set $\mathcal{I}_s^i$ is socially generated and socially biased.

2.5 Key social-friction parameters in Fable 4

The model introduces two transmission-bias parameters.

- **b**: message bias. Each time a signal batch is socially transmitted, its mean is shifted by b. Positive b means optimistic or favorable distortion.
- $\boldsymbol{\kappa}$: meeting-rate misperception. The true Poisson meeting rate is η, but agents think it is $\kappa\eta$. The focal case is $\kappa < 1$: agents underestimate how much others have already learned socially.

Interpretation. Parameter b captures biased content; parameter κ captures biased beliefs about the *social learning process itself*.

2.6 Signal counts, average signals, and bias counts

Let n_s^i be the number of private signals agent i possesses at date s, and let S_{ij}^b denote signal j inclusive of transmission bias. The agent's average private signal is

$$Y_s^i = \frac{1}{n_s^i} \sum_{j=1}^{n_s^i} S_{ij}^b. \tag{6}$$

The precision-weighted average of public signals through date t is

$$Z_t = \frac{\sum_{r=0}^{t} \tau_r^P S_r^P}{\sum_{r=0}^{t} \tau_r^P}. \tag{7}$$

At the population level, the per capita number of signals is

$$\phi_s = \sum_{n=1}^{\infty} n \mu_s(n), \qquad \phi_s^R = \sum_{n=1}^{\infty} n \mu_s^R(n), \tag{8}$$

where μ_s is the distribution perceived by agents and μ_s^R is the actual cross-sectional distribution. The economy-wide per-signal bias count is

$$f_s^b = \frac{\sum_{n_b=1}^{\infty} n_b \nu_s(n_b)}{\phi_s^R}, \tag{9}$$

where $\nu_s(n_b)$ is the cross-sectional frequency of investors with n_b accumulated bias units.

Interpretation. The model separates *how much information exists* from *how biased that information has become.* This distinction is what generates recursive amplification.

2.7 Key objects in Fable 5: folk models, infection, recovery, and buzz

The final fable uses a compartmental approach. A **folk model** is a simplified mental representation of how the world works. Let

- $I(s)$: fraction of agents infected with a folk model,
- $R(s)$: fraction resistant to it,
- $S(s)$: fraction susceptible.

In the SIRS example with an asset market and buzz, price is proportional to the infected fraction,

$$v_s = DI(s), \qquad D > 0, \tag{10}$$

and infection dynamics are modified so that contagiousness depends on $\dot{I}(s)$, the speed with which belief adoption is spreading.

Interpretation. "Buzz" means that a belief becomes more persuasive precisely when it is spreading rapidly. This adds another layer of endogenous positive feedback.

3 Models and theoretical results (all equations + interpretation)

3.1 Fable 1: bandwidth constraints and simplistic thinking

The mechanism is verbal rather than a closed-form system in the published address. Communication bandwidth and limited cognition reduce nuance in transmitted messages. If receivers fail to adjust for the loss of nuance, they infer that senders hold overly simple or extreme views. As receivers become senders, the simplification can iterate.

Interpretation. Society can drift toward oversimplified thinking even if no individual has an intrinsic taste for extremism. The distortion comes from repeated compression of ideas.

3.2 Fable 2: self-enhancing transmission bias

Again, the address presents the mechanism more than a complete closed-form system. Investors choose between two strategies, active A and passive P. The sender in a meeting is more likely to report a high return than a low return. This creates an upward selection bias in the set of returns observed by receivers.

Mechanism.

- High-variance strategies generate more extreme good outcomes.
- If good outcomes are selectively reported, those strategies get discussed more.
- If receivers naively infer skill from reported performance, high-variance strategies spread even when inferior.
- If attention is drawn especially to extreme outcomes, positive skewness spreads for similar reasons.

Interpretation. Apparent attraction to volatility or skewness can be socially emergent rather than a primitive lottery preference.

3.3 Fable 3: visibility bias and overconsumption

This fable is also mostly verbal in the address. Let x denote the need for saving, such as the probability of a personal wealth disaster. Individuals can learn about x from social observation. But engaging in a consumption activity is more visible than not consuming.

Mechanism.

- People observe consumption events more than nonconsumption events.
- If they neglect that visibility bias, they infer that others are consuming heavily.
- They therefore infer that the need for precautionary saving is low.
- Equilibrium overconsumption and undersaving result.

Interpretation. The model generates undersaving without assuming present-biased preferences. The distortion is social learning from selectively visible behavior.

3.4 Fable 4: biased information percolation

3.4.1 Percolation dynamics and recursive bias

The paper derives the evolution of signal counts and bias counts under pairwise meetings. The central proposition is:

Proposition 1. At date s,

$$\phi_s = e^{\eta\kappa s}, \tag{11}$$
$$\phi_s^R = e^{\eta s}, \tag{12}$$
$$f_s^b = \eta s, \tag{13}$$

and the perceived cross-sectional distribution of signal counts is

$$\mu_s(n) = \frac{(\eta\kappa s)^{\log_2 n} e^{-\eta\kappa s}}{(\log_2 n)!}, \qquad n \in \{2^N\}_{N=0,1,\ldots}. \tag{14}$$

Interpretation. The actual number of signals in the economy grows exponentially, but the per-signal bias count grows linearly. Since biases are repeatedly layered onto increasingly large information sets, the total amount of distortion can become very large.

3.4.2 Bayesian-style belief updating with biased private signals

Define the precision weights

$$\lambda^X(n, \tau^S, \tau^X, \tau^Z) = \frac{\tau^X}{\tau^X + n\tau^S + \tau^Z}, \tag{15}$$

$$\lambda^P(n, \tau^S, \tau^X, \tau^Z) = \frac{\tau^Z}{\tau^X + n\tau^S + \tau^Z}, \tag{16}$$

$$\lambda^S(n, \tau^S, \tau^X, \tau^Z) = \frac{n\tau^S}{\tau^X + n\tau^S + \tau^Z}. \tag{17}$$

An agent who ignores transmission bias perceives expected payoff as

$$\mathbb{E}^B[X \mid Y, n; Z, \tau^Z] = \lambda^X \bar{X} + \lambda^S Y + \lambda^P Z, \tag{18}$$

where superscript B means "believed by the biased agent."

In the Action Booms setting, the agent's actual position becomes

$$\theta^i = \frac{\tau^X(\bar{X} - v) + n\tau^S(\bar{X} + \bar{\epsilon}^i + bf_s^{b,i} - v) + \tau^Z(Z - v)}{\gamma}. \tag{19}$$

Interpretation. The optimization problem itself is standard. What changes is the content of the average private signal Y: it contains recursively amplified social bias.

3.4.3 Aggregate action and action booms

Aggregating individual positions yields

$$\Theta_s = \frac{\tau^X(\bar{X} - v) + \phi_s^R \tau^S(\bar{X} + bf_s^b - v) + \tau_s^Z(Z_s - v)}{\gamma}. \tag{20}$$

Taking the rational expectation conditional on b,

$$\mathbb{E}^R[\Theta_s \mid b] = \frac{(\tau^X + \tau_s^Z + \phi_s^R \tau^S)(\bar{X} - v) + \phi_s^R \tau^S b f_s^b}{\gamma}. \tag{21}$$

The excess action relative to the rational benchmark $b = 0$ is

$$\widetilde{\Theta}_s(b) = \frac{\mathbb{E}^R[\Theta_s \mid b] - \mathbb{E}^R[\Theta_s \mid b = 0]}{\mathbb{E}^R[\Theta_s \mid b = 0]} = \frac{\phi_s^R \tau^S b f_s^b}{(\tau^X + \phi_s^R \tau^S + \tau_s^Z)(\bar{X} - v)}. \tag{22}$$

Substituting Proposition 1 and the public precision path gives

$$\widetilde{\Theta}_s(b) = \frac{e^{\eta s} \tau^S b \eta s}{(\tau^X + e^{\eta s} \tau^S + e^{\zeta \lfloor s \rfloor} \tau^P)(\bar{X} - v)}. \tag{23}$$

Interpretation.

- When $b > 0$, social interaction generates overoptimism and an action boom.
- The boom is initially convex because both signal accumulation and per-signal bias are rising.
- Once public information becomes precise enough, the boom corrects.

Proposition 2. Aggregate action follows a hump-shaped path when $b > 0$ and a U-shaped path when $b < 0$.

Proposition 3. More sociable agents (higher meeting rate than the population) take more extreme actions on average: above the per-capita action when $b > 0$, and below it when $b < 0$.

Interpretation. This is a distinctive social prediction. Bubble behavior should be stronger for more socially connected agents.

3.4.4 The Price Bubbles Model: price as a misperceived signal

In the asset-market version, risky-asset supply noise is Gaussian,

$$\Theta \sim N\left(\bar{\Theta}, \frac{1}{\tau^{\Theta}}\right). \tag{24}$$

The rational equilibrium price is conjectured to be

$$v_s = \alpha_s^{RX} X + \alpha_s^{RP} Z_s + \alpha_s^{R\Theta}\Theta + \alpha_s^{Rb} b f_s^b, \tag{25}$$

whereas agents incorrectly perceive

$$v_s \overset{B}{=} \alpha_s^X X + \alpha_s^P Z_s + \alpha_s^{\Theta}\Theta. \tag{26}$$

This lets investors treat normalized price as an extra signal,

$$\xi_s \equiv \frac{v_s}{\alpha_s^X} - \frac{\alpha_s^P}{\alpha_s^X} Z_s \overset{B}{=} X + \frac{\alpha_s^{\Theta}}{\alpha_s^X}\Theta. \tag{27}$$

Perceived conditional variance is

$$o(n, \tau^S, \tau^X, \tau^Z, \tau^{\xi}) \equiv \left(\tau^X + n\tau^S + \tau^Z + \tau^{\xi}\right)^{-1}. \tag{28}$$

Interpretation. Price is informative in reality, but agents misread how informative it is because they underestimate the amount of prior social communication embodied in price.

3.4.5 Perceived pricing coefficients and actual market price

The perceived pricing coefficients are

$$\alpha_s^X = \frac{\phi_s \tau^S(\gamma^2 + \phi_s \tau^S \tau^\Theta)}{(\phi_s \tau^S)^2 \tau^\Theta + \gamma^2(\tau^X + \phi_s \tau^S + \tau_s^Z)}, \tag{29}$$

$$\alpha_s^\Theta = -\frac{\gamma(\gamma^2 + \phi_s \tau^S \tau^\Theta)}{(\phi_s \tau^S)^2 \tau^\Theta + \gamma^2(\tau^X + \phi_s \tau^S + \tau_s^Z)}, \tag{30}$$

$$\alpha_s^P = \frac{\gamma^2 \tau_s^Z}{(\phi_s \tau^S)^2 \tau^\Theta + \gamma^2(\tau^X + \phi_s \tau^S + \tau_s^Z)}. \tag{31}$$

The actual equilibrium price is

$$v_s(X, Z_s, \Theta, b) = \frac{\tau_s^Z \gamma^2 Z_s + (\gamma^2 + \phi_s \tau^S \tau^\Theta)\left[\phi_s^R \tau^S (X + b f_s^b) - \gamma \Theta\right]}{\gamma^2(\tau^X + \phi_s^R \tau^S + \tau_s^Z) + \phi_s \phi_s^R (\tau^S)^2 \tau^\Theta}. \tag{32}$$

Proposition 4. The perceived price coefficients depend on ϕ_s, the *perceived* per-capita signal count.

Proposition 5. The actual price depends on ϕ_s^R, the *actual* per-capita signal count, and on the recursively amplified bias term $b f_s^b$.

Proposition 6. If $\kappa < 1$, then investors underweight the information in price: the true coefficient on fundamentals is larger, and the true dependence on public noise and supply noise is smaller, than investors believe.

Interpretation. κ-bias creates excessive willingness to trade against the market because people think price reveals less about others' information than it actually does.

3.4.6 Expected price path, bubble phases, and return anomalies

Taking expectations conditional on b gives

$$\mathbb{E}^R[v_s \mid b] = \frac{(\gamma^2 + \phi_s \tau^S \tau^\Theta)\phi_s^R \tau^S b f_s^b}{\gamma^2(\tau^X + \phi_s^R \tau^S + \tau_s^Z) + \phi_s \phi_s^R (\tau^S)^2 \tau^\Theta}. \tag{33}$$

Interpretation. This expected price path is hump-shaped when $b > 0$: price rises above fundamental value, reaches a peak, and then returns toward fundamental value as public information eventually dominates biased percolation.

The paper emphasizes four phases for $b > 0$:

- **Inception phase:** expected price grows at an accelerating rate.
- **Boom phase:** expected price keeps growing but at a decreasing rate.
- **Correction phase:** expected price falls at an increasing rate.
- **Terminal phase:** expected price continues falling but at a decreasing rate toward fundamental value.

The main dynamic asset-pricing implications are:

- **Proposition 7.** When $b > 0$, prices drop on average at public-news arrival dates because the stock is temporarily overpriced and news partially corrects that overpricing.
- **Proposition 8.** Conditional on b, the sign of long-horizon postevent return continuation matches the sign of the event-date reaction.
- **Proposition 9.** In a mixed sample of news events, variation in b can generate long-horizon continuation after public announcements, consistent with PEAD-like behavior.
- **Proposition 10.** At short horizons, successive announcement-related price changes can continue in the same direction.
- **Proposition 11.** With sufficiently smooth public information and sufficiently large $V(b)$, short-horizon returns are positively autocorrelated.
- **Proposition 12.** Investor expectations become more optimistic after recent price increases, creating the *appearance* of extrapolation even though no agent extrapolates mechanically.

Interpretation. This is the paper's most striking message. Momentum, reversal, PEAD, aggressive trading, and investor optimism can all arise from distorted social learning rather than from direct overconfidence or overextrapolation.

3.5 Fable 5: folk models, SIRS dynamics, and buzz

The final fable adapts a compartmental epidemic model. The law of motion in the example with an asset market and buzz is

$$\dot{R}(s) = rI(s) - kR(s), \tag{34}$$
$$\dot{I}(s) = -rI(s) + m\big(a + bD\dot{I}(s)\big)I(s)\big(1 - I(s)\big), \tag{35}$$
$$v_s = DI(s), \qquad D > 0. \tag{36}$$

Here r is the recovery rate, k is the rate at which resistance decays, m is meeting intensity, a is baseline contagiousness, and b is the buzz parameter.

Interpretation.

- A folk model spreads like an infection.
- "Buzz" means that rapid current adoption makes the model even more contagious.
- Price rises when infected believers become more common.
- Correction can overshoot, producing damped oscillations and even a dead-cat-bounce type recovery after the crash.

Economic meaning. Folk models are not just colorful stories. They are socially contagious state variables that can move expectations, trading, and prices.

4 Identification, discipline, and empirical implications

4.1 What disciplines the theoretical argument

The paper is broad in vision, but it does not rely on vague storytelling alone. The formal core is disciplined by:

- standard Gaussian signals and Bayesian-style updating,
- mean-variance demand under CARA utility,
- explicit market clearing in the price-bubbles model,
- explicit dynamics for meetings, signal accumulation, and bias accumulation,
- public signals that eventually become informative enough to correct mispricing.

Interpretation. The paper is not saying that "anything social can happen." It shows, in a highly structured environment, exactly how biased communication changes equilibrium outcomes.

4.2 Why the social mechanism is different from direct behavioral bias

The address repeatedly stresses **emergence** and **mimicry**.

- High variance and skewness can spread without lottery preferences.
- Overconsumption can arise without present-biased utility.
- Beliefs can look extrapolative without any direct extrapolation rule.
- Bubbles can arise without requiring strong direct overconfidence.

Interpretation. Observed behavior does not reveal primitive psychology one-for-one. Social systems can transform mild conversational biases into large aggregate outcomes.

4.3 Distinctive empirical implications

The paper suggests several empirical margins that are especially diagnostic.

- **Sociability and network centrality.** More social agents should show stronger boom behavior and more extreme positions.
- **Communication technology.** Blogs, social media, and electronic communication may alter both the amount of interaction and the form of transmission bias.
- **Sender incentives and personality.** Self-enhancing transmission should be stronger for people with stronger self-presentation motives.
- **Visibility.** Highly visible consumption categories should be especially prone to social overconsumption effects.
- **Message content.** Narrative vividness, positivity/negativity, and simplification should matter for how beliefs spread.

Interpretation. The model implies that finance researchers should not only measure firm characteristics or investor demographics. They should also measure social connectedness, media environments, and transmission-bias proxies.

4.4 Policy implications

Because the distortions in Fables 2 and 3 operate through biased social learning, policy implications differ sharply from standard behavioral models.

- Better disclosure can reduce overconsumption when the problem is biased inference from visible behavior.
- Interventions targeted at socially central individuals may be more effective than uniform interventions.
- Regulation of information environments may matter not just because of the amount of information, but because of how information is selectively amplified.

Interpretation. Social structure is a policy lever. The same subsidy or disclosure can work very differently depending on who talks to whom and how messages are distorted.

4.5 Limitations and caveats

The paper is honest about its limits.

- Several fables are sketches or work in progress rather than complete models.
- Fable 4 assumes homophily by signal count, myopic optimization, and forgetting of past prices.
- The baseline price model does not include rational arbitrageurs who fully understand b-bias and κ-bias.
- Fable 5 uses a deliberately stylized SIRS example with deterministic price dynamics.
- Social interaction can improve collective intelligence as well as degrade it; the paper emphasizes the neglected downside channel.

Interpretation. These are not flaws that negate the contribution. They indicate that the paper is opening a research program, not closing one.

4.6 Relation to prior paradigm shifts

The address frames social economics and finance as a next step after two earlier paradigm shifts.

- **Information economics:** people possess different private information.
- **Behavioral economics and finance:** people make systematic mistakes.
- **Social economics and finance:** people observe and talk to each other, and those social processes systematically reshape ideas.

Interpretation. The paper complements rational social-learning and network models by emphasizing not just *who is linked to whom*, but *how messages are biased as they travel.*

5 Figures and key theoretical results

5.1 Figure 2: six themes about social economics and finance

Six Themes about Social Economics and Finance

I offer six themes about a new intellectual paradigm, *social economics and finance*, and one of its key intellectual building blocks, *social transmission bias*.

- **Compounding:** Social transmission bias compounds recursively, so small bias can have large effects.
- **Idiosyncrasy:** Social transmission bias helps explain why aggregate outcomes are often error-prone and unpredictable.
- **Dynamics:** Social transmission bias offers an endogenous *social* explanation for action booms, price bubbles, and swings in investor sentiment.
- **Emergence:** Socially emergent behavior often looks completely different from individual propensities.
- **Mimicry:** Social emergence can easily create the illusion of a direct individual propensity "for" a behavior when no such propensity exists. So the inferences drawn from empirical tests of behavioral hypotheses are often overstated.
- **Proxies:** This approach suggests *new test variables* for empirical research: (i) general social and network proxies, and (ii) proxies for transmission bias.

Figure 2. Six themes summarized by Hirshleifer.

Read:

- Small transmission bias can compound into large aggregate effects.
- Social outcomes can be highly idiosyncratic and hard to predict.
- Social processes help explain booms, bubbles, and sentiment swings.
- Emergent behavior can look unlike any primitive individual taste or bias.
- Apparent direct biases can be mimicry generated by social transmission.
- The theory points to new empirical proxies based on networks and communication environments.

Economic message. This figure is the paper in one page. Hirshleifer is proposing a new research agenda, not just a single model.

5.2 Figure 4: timeline of the basic percolation model

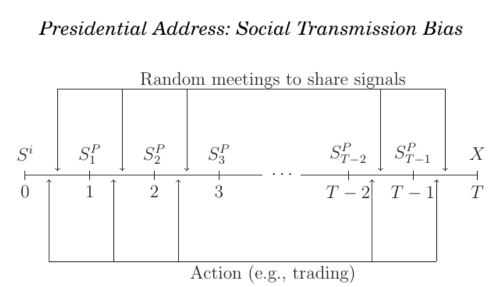

Figure 4. Timeline for actions, private signals, public signals, and final payoff realization.

Read:

- Each agent starts with one private signal at date 0.
- Random social meetings happen in continuous time.
- Public signals arrive at discrete dates and become progressively more informative.
- Agents choose actions continuously before the common payoff is realized at T.

Economic message. The bubble mechanism is not driven by pure psychology in a vacuum. It is driven by the interaction of repeated social meetings with a gradually revealing public-information process.

5.3 Figure 3: expected price path in the Price Bubbles Model

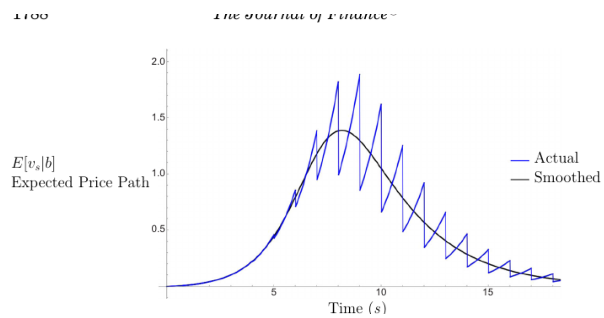

Figure 3. Expected price path with positive transmission bias $b > 0$.

Read:

- The smoothed path is hump-shaped: inception, boom, correction, terminal phase.
- The unsmoothed path zigzags because public signals partially correct mispricing at discrete dates.
- Oscillation is strongest near the peak, when biased percolation and public correction temporarily have comparable force.
- Price eventually returns toward fundamental value once public precision dominates.

Economic message. This is the paper's signature result: recursive message bias alone can generate a bubble that looks "beyond all reason" before collapsing.

5.4 Figure 5: SIRS folk-model contagion with buzz

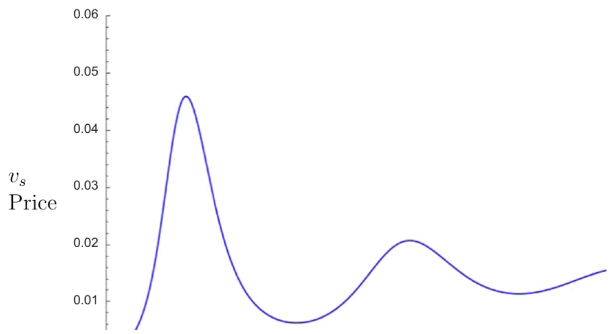

Figure 5. Price path in the SIRS folk-model example with buzz.

Read:

- Price rises sharply when a folk model spreads through the population.
- Recovery from the model can overshoot, producing damped oscillations.
- Buzz amplifies both the boom on the way up and the collapse on the way down.
- The postcrash path can include a dead-cat-bounce type rebound.

Economic message. Narrative contagion and belief epidemics can produce bubble dynamics even outside the fully information-based percolation model.

5.5 Summary of key theoretical results

Block	Formal content	Economic meaning
Proposition 1	$\phi_s^R = e^{\eta s}$ and $f_s^b = \eta s$.	Information and transmission bias both accumulate over time; recursive amplification is built into the state variables.
Propositions 2–3	Aggregate action follows hump/U shapes; more sociable agents act more extremely.	Social connectedness is not incidental—it is a state variable that predicts bubble exposure.
Propositions 4–6	Investors misread price because they misperceive how much information others have already socially aggregated.	κ-bias creates excessive trading and mispricing by making price look less informative than it really is.
Propositions 7–10	Public news only partially corrects bubbles, so event-date reactions and postevent returns are predictably linked.	The model generates PEAD-like continuation and short-horizon continuation from biased social learning rather than direct underreaction to fundamentals.
Propositions 11–12	Short-run returns can be positively autocorrelated, and optimism rises after price run-ups.	Momentum and extrapolation-like patterns can be socially emergent mimicry rather than primitive investor rules.

Table 1. Grouped summary of the main formal results in Table 4.

6 Short summary

- Hirshleifer proposes social economics and finance as a new paradigm for understanding how social interaction shapes beliefs and behavior.
- The key building block is social transmission bias: signals or ideas are systematically distorted or selectively transmitted in social contact.
- Fable 1 shows how bandwidth constraints and limited cognition can generate simplistic thinking.
- Fable 2 shows how self-enhancing transmission can make high-variance or high-skewness strategies spread without direct lottery preferences.
- Fable 3 shows how visibility bias can generate overconsumption and undersaving without present-biased utility.
- Fable 4 provides a formal model of biased information percolation that generates action booms, price bubbles, trading surges, event-based return continuation, momentum-like patterns, and optimism that mimics extrapolation.
- Fable 5 shows how folk models and narratives can spread through SIRS-style contagion with buzz, generating booms, crashes, and damped oscillations.
- The paper's broad lesson is that many apparently individual-level biases may actually be emergent outcomes of biased social communication.
- Empirically, the framework points researchers toward proxies for sociability, network position, communication technologies, visibility, and transmission-bias sources.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

Hirshleifer argues that finance and economics need to treat social transmission as a first-order force. When signals pass from person to person with distortion or selective transmission, mild individual-level biases can compound into large social and market outcomes. The paper's formal centerpiece shows that recursive communication bias can generate booms, bubbles, oscillation, trading anomalies, and belief dynamics that look very similar to familiar behavioral-finance patterns.

7.2 Economic intuition

The intuition is recursive amplification. A slightly biased message changes one person's belief. That person then becomes a sender, so the bias is passed on again, often with further distortion. At the same time, receivers typically fail to correct fully for the social process that generated what they observed. Because the bias is multiplied by repeated meetings, it can outrun fundamentals for a while. Only when strong corrective forces—especially informative public news—arrive does the boom reverse.

7.3 Takeaway

The paper's lasting contribution is conceptual: many economic and financial phenomena should be re-read through a social lens. Instead of asking only whether investors are overconfident, extrapolative, impatient, or lottery-seeking, we should also ask whether these behaviors are *socially emergent consequences of transmission bias.* That shift in viewpoint is the essence of Hirshleifer's proposed paradigm of social economics and finance.